

# STEP I - 1987

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## **Abstract**

A collection of solutions to STEP questions for my own use and also for former colleagues and friends. These are very much incomplete. I aim to complete some sporadically although I make absolutely zero promise of doing so with any sort of regular schedule. I largely do this for fun. Although I try my best, I have undoubtedly made several errors so please do use with caution. Any errors, please do let me know at [ccbarnesni@gmail.com](mailto:ccbarnesni@gmail.com).

1. Calculate the derivative:

$$f'(x) = ae^{ax} \cos bx - be^{ax} \sin bx$$

This vanishes when

$$\begin{aligned} ae^{ax} \cos bx &= be^{ax} \sin bx \\ \implies \tan bx &= \frac{b}{a} \quad (\text{since } e^{ax} \neq 0) \\ \implies x_k &= \frac{1}{b} \left( \arctan \left( \frac{a}{b} \right) + k\pi \right), \quad k \in \mathbb{Z} \end{aligned}$$

are the stationary points of  $f$ .

Before finding  $f(x_k)$ , note that  $\cos bx_k = \frac{(-1)^k b}{\sqrt{a^2 + b^2}}$  by Pythagoras' theorem. Note that  $a, b > 0$  implies  $\arctan \left( \frac{b}{a} \right)$  lies in the 1st quadrant. Then  $bx_k$  must alternate between the quadrants I and III. Then,

$$f(x_k) = \frac{(-1)^k ab}{\sqrt{a^2 + b^2}} \exp \left( \frac{a}{b} \left( \arctan \left( \frac{a}{b} \right) + k\pi \right) \right)$$

from which it follows that  $f(x_k)$  forms a geometric progression with common ratio  $-e^{\frac{a\pi}{b}}$ . A sketch is below.

2.

3. Substitute  $v(x) = y(x)/x$  into the differential equation to obtain

$$\begin{aligned} x^3 \frac{d}{dx} \left( \frac{y}{x} \right) + xy &= \frac{x(1+y^2)}{y(1+x^2)} \\ x^2 \frac{dy}{dx} - xy + xy &= \frac{x(1+y^2)}{y(1+x^2)} \\ x^2 \frac{dy}{dx} &= \frac{x(1+y^2)}{y(1+x^2)} \\ \frac{y}{1+y^2} \frac{dy}{dx} &= \frac{1}{x(1+x^2)} \\ \int \frac{y}{1+y^2} dy &= \int \frac{1}{x(1+x^2)} dx \\ \int \frac{y}{1+y^2} dy &= \int \frac{1}{x} - \frac{x}{1+x^2} dx \\ \frac{1}{2} \ln |1+y^2| + c &= \ln |x| - \frac{1}{2} \ln |1+x^2| \end{aligned}$$

Using the initial conditions  $v(1) = 1$  we find that  $y(1) = 1$  and so

$$c = -\ln 2$$

Grouping the logs together, we have

$$\begin{aligned}\ln \sqrt{(1+y^2)(1+x^2)} &= \ln 2x \\ (1+y^2)(1+x^2) &= 4x^2 \\ y^2 &= \frac{3x^2-1}{1+x^2} \\ v^2 x^2 &= \frac{3x^2-1}{1+x^2} \\ v^2 &= \frac{3x^2-1}{x^2(1+x^2)}\end{aligned}$$

From above, we see that  $v \rightarrow 0$  as  $x \rightarrow \infty$ .

*This doesn't have a real solution for  $x \in (-1/\sqrt{3}, 1/\sqrt{3})$ , is this definitely correct?*

4. First observe that we can change the base on the  $n^{th}$  term as follows:

$$(-1)^n \log_{2^{2^n}} e = \frac{(-1)^n \ln e}{2^n \ln 2} = \frac{(-1)^n}{2^n \ln 2}$$

It is readily seen that this is a geometric series with first term  $\frac{1}{\ln 2}$  and common ratio  $-\frac{1}{2}$ . The infinite sum is thus

$$\frac{1/\ln 2}{1+1/2} = \frac{1}{(3/2)\ln 2} = \frac{1}{\ln 2\sqrt{2}}$$

5. First note the change of limits. If  $x = \alpha$  then  $\theta = 0$ . Similarly,  $x = \beta \implies \theta = \pi/2$ . Then

$$\frac{dx}{d\theta} = 2(\beta - \alpha) \sin \theta \cos \theta.$$

Also,

$$x - \alpha = \alpha(\cos^2 \theta - 1) + \beta \sin^2 \theta = (\beta - \alpha) \sin^2 \theta,$$

and similarly

$$\beta - x = \beta(1 - \sin^2 \theta) - \alpha \cos^2 \theta = (\beta - \alpha) \cos^2 \theta.$$

The integral thus becomes

$$\begin{aligned}& \int_0^{\pi/2} \frac{2(\beta - \alpha) \sin \theta \cos \theta}{\sqrt{(\beta - \alpha)^2 \sin^2 \theta \cos^2 \theta}} d\theta \\ &= 2 \int_0^{\pi/2} d\theta \\ &= \pi\end{aligned}$$

In the case where  $\alpha > \beta$  the integral becomes

$$- \int_0^{\pi/2} \frac{2(\beta - \alpha) \sin \theta \cos \theta}{\sqrt{(\beta - \alpha)^2 \sin^2 \theta \cos^2 \theta}} d\theta$$

which is also  $\pi$ . To see this, first we obtain the first negative by swapping the limits of integration using the result that  $\int_{\alpha}^{\beta} dx = -\int_{\beta}^{\alpha} dx$ . Then note that  $\beta - \alpha$  is now a negative quantity also however this does not cancel with the denominator since the presence of the radical ensures the denominator is, in fact, positive root. The trigonometric functions are all positive in the interval  $[0, \frac{\pi}{2}]$  so all in all we have two additional negatives. The answer is thus  $\pi$ .

For the second integral, performing the same substitution results in

$$\int_0^{\frac{\pi}{2}} \frac{1}{\alpha \cos^2 \theta + \beta \sin^2 \theta} d\theta$$

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15. Let  $\Theta \sim U(2\pi)$  be the random variable denoting the angle between the line segment connecting the random point on the unit circle to the original and the positive  $x$ -axis. It follows then that  $X = \sqrt{(1 - \cos \theta)^2 + \sin^2 \theta}$ . Note that we must also have  $\mathbb{P}(a < X < b) = 2\mathbb{P}(\theta_1 < \Theta < \theta_2)$  since there are two intervals along the unit circle for which the random variable  $X$  takes the same value. Also, for a given  $\theta$ , using the cosine rule one can show that the corresponding distance from  $(\cos \theta, \sin \theta)$  to  $(1, 0)$  must satisfy  $\theta = \arccos(\frac{2-x^2}{2})$ . Then, putting this together we can determine the CDF of the  $X$  as follows.

$$\begin{aligned} \mathbb{P}(a < X < b) &= 2\mathbb{P}(\theta_1 < \Theta < \theta_2) \\ &= \frac{\theta_2 - \theta_1}{\pi} \\ &= \frac{\arccos(\frac{2-b^2}{2}) - \arccos(\frac{2-a^2}{2})}{\pi} \\ &= \frac{1}{\pi} [\arccos(\frac{2-x^2}{2})]_a^b \\ &= \frac{1}{\pi} \int_a^b \frac{x}{\sqrt{1 - (\frac{2-x^2}{2})^2}} dx \\ &= \frac{1}{\pi} \int_a^b \frac{2}{\sqrt{4 - x^2}} dx \end{aligned}$$

We thus compute the mean

$$\begin{aligned}\mathbb{E}(X) &= \frac{2}{\pi} \int_0^2 \frac{x}{\sqrt{4-x^2}} dx \\ &= -\frac{2}{\pi} \left[ (4-x^2)^{\frac{1}{2}} \right]_0^2 \\ &= \frac{4}{\pi}.\end{aligned}$$

The 2nd moment is calculated using the substitution  $x = 2 \sin \theta$  as

$$\begin{aligned}\mathbb{E}(X^2) &= \frac{2}{\pi} \int_0^2 \frac{x^2}{\sqrt{4-x^2}} dx \\ &= \frac{8}{\pi} \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta \\ &= \frac{4}{\pi} \int_0^{\frac{\pi}{2}} 1 - \cos 2\theta d\theta \\ &= \frac{4}{\pi} \left[ \theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}} \\ &= 2\end{aligned}$$

Thus the variance is

$$\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}^2(X) = 2 - \frac{16}{\pi^2}$$

The probability that  $X$  is greater than its mean is thus

$$\begin{aligned}\mathbb{P}(X > \mu) &= \mathbb{P}\left(X > \frac{4}{\pi}\right) \\ &= \mathbb{P}\left(\frac{4}{\pi} < X \leq 2\right) \\ &= \frac{2}{\pi} \int_{\frac{4}{\pi}}^2 \frac{1}{\sqrt{4-x^2}} dx \\ &= \frac{2}{\pi} \int_{\arcsin\left(\frac{2}{\pi}\right)}^{\frac{\pi}{2}} d\theta \\ &= 1 - \frac{2}{\pi} \arcsin\left(\frac{2}{\pi}\right)\end{aligned}$$

16.